arc_temp_humid - outstanding work

Updated 6 September 2026. The repository rename and the main-site move are complete and are no longer listed here; see `CHANGELOG.md` for what was done.

Everything described below is **not yet done**. What *is* done is in `CHANGELOG.md`; the architecture is in `CLAUDE.md`.

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1. UK overheating threshold - RESOLVED, no band applies

Both UK datasets keep `"threshold": None`. That is now a decision rather than a placeholder.

CIBSE TM59 (2017) section 4.2 requires **both** of:

- **(a) Living rooms, kitchens and bedrooms.** Hours where dT is at least 1 K, May to September, no more than 3% of occupied hours. dT is measured against the **adaptive** limit (TM52 Criterion 1), not a fixed temperature.
- **(b) Bedrooms only.** Operative temperature between 22:00 and 07:00 not above **26 C** for more than 1% of annual hours - TM59 counts that as 32 hours, so 33 or more is a fail.

The only fixed figure in TM59 is that 26 C, and it applies to bedrooms at night. Both UK sensors are in **living rooms** (Grove `169502D1`, and Holywell `0E3C12EC`, confirmed and renamed from "Internal Ambient"), so criterion (b) does not apply to either and there is no fixed band to draw. Criterion (a) is the adaptive test, which the comfort chart already serves.

If a bedroom sensor is added later, the threshold is currently per-region, so a per-logger threshold would be needed to draw 26 C for that sensor alone.

Two caveats before quoting TM59 numbers from this dashboard

- TM59 is a **design and modelling** methodology: standard occupancy profiles, dynamic simulation, and **operative** temperature. This dashboard plots **measured air**

temperature, already listed as a known limitation in `adaptivecomfort.md`. Numbers from here are indicative of TM59, not a TM59 assessment.
- Criterion (a) is adaptive against **TM52 Category II** (`0.33 x Trm + 18.8 + 3`), which is not a band the dashboard currently offers. See section 2.

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2. TM52 / EN 16798-1 bands - DECIDED, not adding them

No TM59 claims are intended, so the UK datasets stay on **ASHRAE 55 80%** and no TM52 bands are added.

Worth recording why that is a safe default rather than merely a convenient one: across the UK running-mean range the TM52 Cat II upper limit sits about 0.7 to 0.9 K **above** ASHRAE's, so ASHRAE is the more conservative of the two. Nothing is being flattered by the choice.

Upper limit at	Trm 10 C	Trm 15 C	Trm 20 C
ASHRAE 55 80% (in use)	24.4	26.0	27.5
TM52 / EN 16798-1 Cat II	25.1	26.8	28.4
TM52 / EN 16798-1 Cat III	26.1	27.8	29.4

On the current data the choice changes nothing anyway: TM59 criterion (a) exceedance is 0.00% at Grove under every model, and at Holywell 0.53% on ASHRAE against 0.27% on Cat II, both far inside the 3% allowance.

If that ever changes and a TM59 result does need stating, the line has to be the one TM59 names. Adding TM52 Cat II and Cat III is two entries in `COMFORT_MODELS` plus labels. Note that EN 16798-1 designates **Category III for existing buildings**, which these retrofits are, so Cat III would likely be the right one rather than Cat II.

3. Open-Meteo coordinates and date ranges - DONE

All three feeds now publish a coordinate that is not the building, and start from the building's own record rather than an arbitrary date.

Feed	Committed coordinate	Distance from building	Feed starts
`tz`	`-7.07, 39.30`	0.56 km	2023-02-12

Feed	Committed coordinate	Distance from building	Feed starts
`grove`	`52.057774, -2.704697`	0.60 km	2026-07-01
`holywell`	`52.926643, -4.230377`	0.65 km	2026-07-01

The UK values are Open-Meteo model grid cell centres. Requesting any point inside a cell returns that cell's series, so this is byte-identical data while the published number is a fixed feature of the weather model rather than a private address. The town-centre coordinates used before were already in the same cells, so the weather data was correct all along - what changed is what gets published.

Tanzania is rounded to 2dp instead, because Open-Meteo serves that region from a coarser global model and echoes the requested point rather than snapping to a grid. Rounding costs nothing there: even 1dp (3.9 km away) returns byte-identical data. The previous value carried seven decimals, which is centimetre-level precision on a children's facility in a public repository.

The 30-day lead-in, and a bug it fixed

Each `start_date` is the building's first sensor reading minus 30 days. The lead-in is not padding: the EN 16798-1 running mean is seeded from its own first day and decays that seed by alpha = 0.8 daily, so without a run-up the opening fortnight of comfort points are measured against a running mean still carrying an arbitrary starting value. Thirty days reduces the seed's weight to about 0.1%.

This turned out to matter for Tanzania, not just the UK. House 5's first sensor reading is **2023-03-14**, while the Open-Meteo feed began **2023-03-15** - a day *after* the data it exists to contextualise. Every House 5 comfort point in the first couple of weeks of the record was being measured against a running mean with no history behind it. The feed now starts 2023-02-12.

If sensors are ever installed earlier than the current start dates, move `start_date` back to 30 days before the new earliest reading.

4. ARC UK Omnisense fetch - DONE and tested

`SITES["uk"]` uses site number **58345** ("Simmonds.Mills Retrofits") with the shared `OMNISENSE_USERNAME` / `OMNISENSE_PASSWORD` secrets, and a `--site uk` step runs daily in `update-dashboard-data.yml` beside the Tanzanian one.

Exercised in CI on 6 September 2026 via `debug-omnisense.yml`, which now takes a site input so either site can be tested without waiting for the nightly run or committing anything:

```
Omnisense fetch [uk] - 2026-09-06 17:29 UTC
  Date range: 2026-07-31 -> 2026-09-06
  Login successful.
  Download ready: 180855 rows
  Wrote 10.3 MB -> data/omnisense_uk/omnisense_uk_20260906_1729.csv
```

So the shared credentials do reach the UK site, as expected from the export containing a House 5 sensor alongside the Grove and Holywell ones.

To re-test at any point: Actions -> Debug Omnisense fetch -> Run workflow -> site `uk`. It downloads and discards, committing nothing.

5. Add the UK feeds to the staleness monitor

`check_staleness.py` watches the Tanzanian Omnisense feed and Open-Meteo, and drives both `data/status.json` and the email alerts. It does **not** yet know about `omnisense_uk`, `openmeteo_grove` or `openmeteo_holywell`.

The consequence showed up on 7 September 2026: the UK Omnisense fetch failed and nothing raised it. The dashboard's own sidebar does show "Omnisense (UK) last updated", so it is visible to anyone looking, but there is no proactive alert the way there is for Tanzania.

Adding them means an entry per feed in the `THRESHOLDS` / `LABELS` dicts near the top of `check_staleness.py` and a `sources.append(entry(...))` alongside the existing ones.

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6. Optional: Copernicus climate data for the UK

Long-Term Mode overlays Copernicus ERA5 historic and CMIP6 SSP projection series. Those are drawn for **one user-defined region** in the Copernicus Interactive Climate Atlas, and the set in `data/hist_proj/` is Tanzanian - its ERA5 series averages about 25 C, which is meaningless over Herefordshire.

Long-Term Mode is therefore **hidden entirely on the UK datasets** rather than showing another region's projections under a UK building. `CLIMATE_REGIONS` in `build.py` maps a region key to a folder and a display label, and each dataset names the region that applies to it; a dataset with no region has no Long-Term Mode.

To add the UK:

1. In the Copernicus Interactive Climate Atlas, draw the region and export the ERA5 historic and the CMIP6 SSP timeseries as CSV.
2. Put them in a new folder, e.g. `data/hist_proj_uk/`, with the same filenames (`t-ERA5_timeseries_historic.csv`, `t-CMIP6_timeseries_SSP*.csv`).
3. Add a `CLIMATE_REGIONS` entry pointing at it with a display label.
4. Set `"climate_region"` on the UK datasets.

Grove and Holywell are about 150 km apart, so strictly they warrant separate regions, in the same way their Open-Meteo feeds do. One "ARC UK" region drawn around both would be a reasonable simplification for climate projections, whose grid is far coarser than weather - but that is a judgement worth making deliberately rather than by default.

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7. Known limitation, not a task

ASHRAE 55 Section 5.4.1(a) requires that no heating is in operation. Nothing at any site records heating status, and nothing in a temperature and humidity trace reliably separates a heated room from a merely warm one.

The dashboard states this rather than inferring it: a bolded line in the sidebar applicability panel directly beneath the quoted criterion, and in the tooltips explaining the method. The 10-33.5 C running-mean gate is enforced automatically and greys out readings the standard does not cover.

That gate now has teeth it did not have before, because the UK Open-Meteo feeds reach back to March 2023: once UK winter indoor data exists, cold-weather readings will be greyed and excluded from the comfort percentage. But **a mild day with the heating on will still pass every test the dashboard can apply**.

Closing this properly needs a heating-status input, not a cleverer algorithm. See `adaptivecomfort.md` section 2 for the full reasoning, including why calendar-month filtering was considered and rejected.